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(54) **PBN HEATERS FOR ALD TEMPERATURE UNIFORMITY**

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(56) **References Cited**

U.S. PATENT DOCUMENTS

7,020,537 B2 3/2006 Wilson et al.

7,160,421 B2 1/2007 Wilson et al.

(Continued)

FOREIGN PATENT DOCUMENTS

CN 105514014 A 4/2016

CN 105957820 A 9/2016

(Continued)

OTHER PUBLICATIONS

English translation to JP2008098513 (Year: 2006).*

(Continued)

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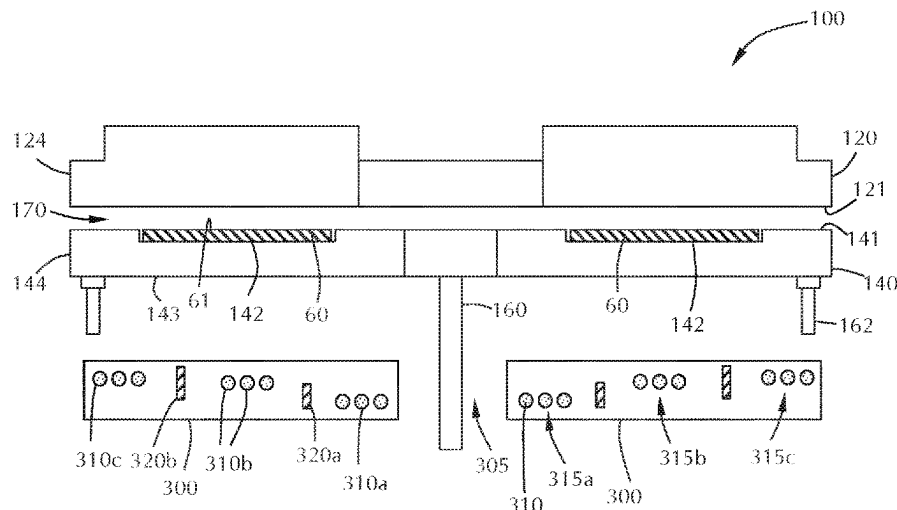
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(57) **ABSTRACT**

Heaters having a body with having a top and bottom comprising pyrolytic boron nitride (PBN), a first heater electrode and a second heater electrode are described. The heater electrodes can be enclosed within an electrically insulating standoff and connected to separate busbars to provide power. Heater assemblies including one or more of the heaters and processing chambers including the heater assemblies are also described.

14 Claims, 7 Drawing Sheets



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USPC 219/444.1, 438
See application file for complete search history.
- (56) **References Cited**
- U.S. PATENT DOCUMENTS
- 7,189,318 B2 3/2007 Wilson et al.
7,751,908 B2 7/2010 Chang et al.
8,399,809 B1 3/2013 Bourez
9,023,664 B2 5/2015 Chang et al.
2002/0108714 A1 8/2002 Doering et al.
2004/0074898 A1* 4/2004 Mariner H01L 21/68757
219/465.1
2007/0131674 A1 6/2007 Kushihashi et al.
2011/0039399 A1 2/2011 Suzuki et al.
2012/0085747 A1* 4/2012 Chao H01L 21/68757
219/444.1
2014/0263281 A1* 9/2014 Kano H05B 3/06
219/544
- 2015/0228513 A1 8/2015 Parkhe et al.
2016/0126125 A1 5/2016 Okugawa et al.
2016/0270150 A1* 9/2016 Furutani H05B 3/26
2019/0177847 A1* 6/2019 Tomizawa C23C 16/4586
2020/0340102 A1* 10/2020 Kimura H01L 21/02274
- FOREIGN PATENT DOCUMENTS
- JP S6214923 A 1/1987
JP H07315935 A 12/1995
JP 2002184558 A 6/2002
JP 2003059789 A 2/2003
JP 2008098513 * 10/2006
JP 2007157661 A 6/2007
JP 2009149964 A 7/2009
JP 2013191626 * 9/2013
JP 2016076646 A 5/2016
JP 2016171066 A 9/2016
KR 20160041758 A 4/2016
KR 20160110130 A 9/2016
WO 2008008585 A2 1/2008
WO 2015183661 A1 12/2015
- OTHER PUBLICATIONS
- English translation to JP 2013191626 (Year: 2013).
PCT International Search Report and Written Opinion in PCT/US2019/018668 dated Jun. 5, 2019, 13 pages.
- * cited by examiner

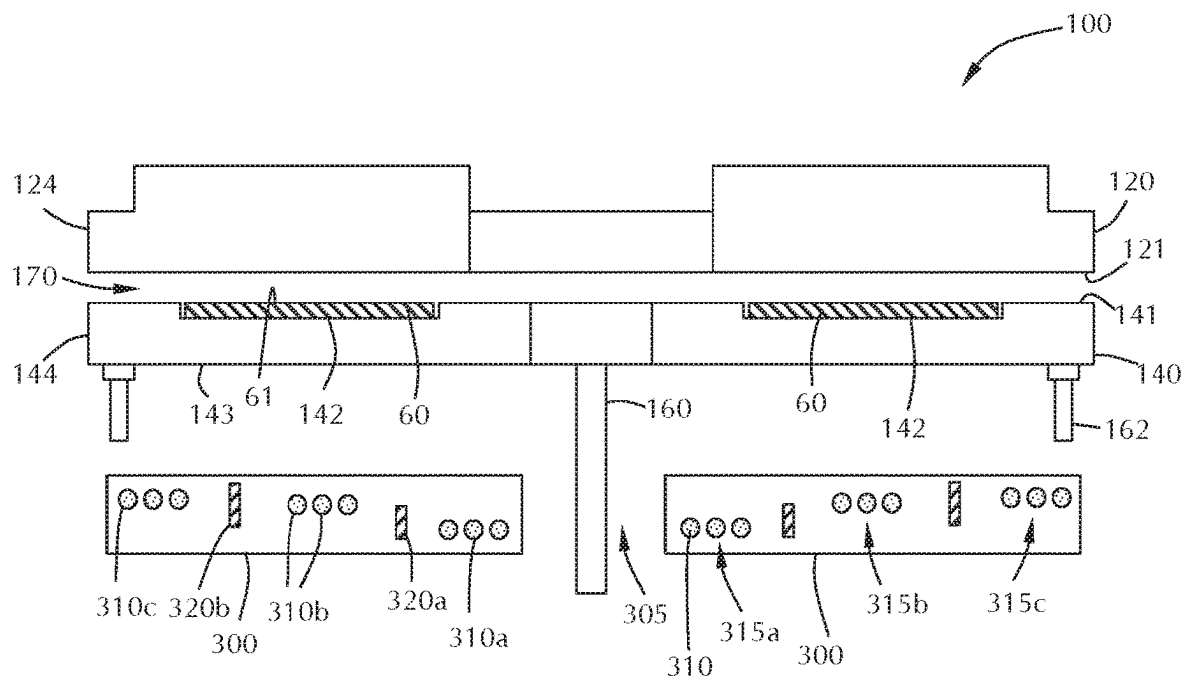


FIG. 1

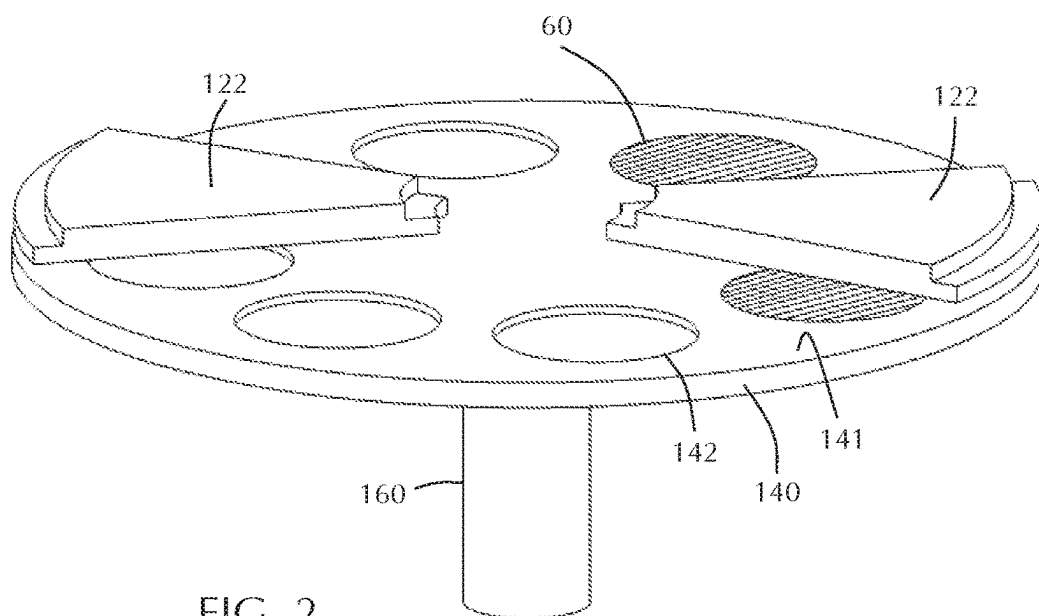


FIG. 2

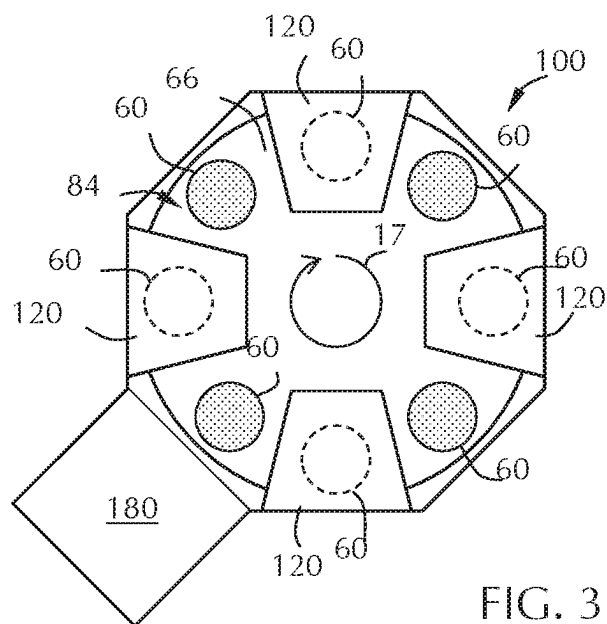


FIG. 3

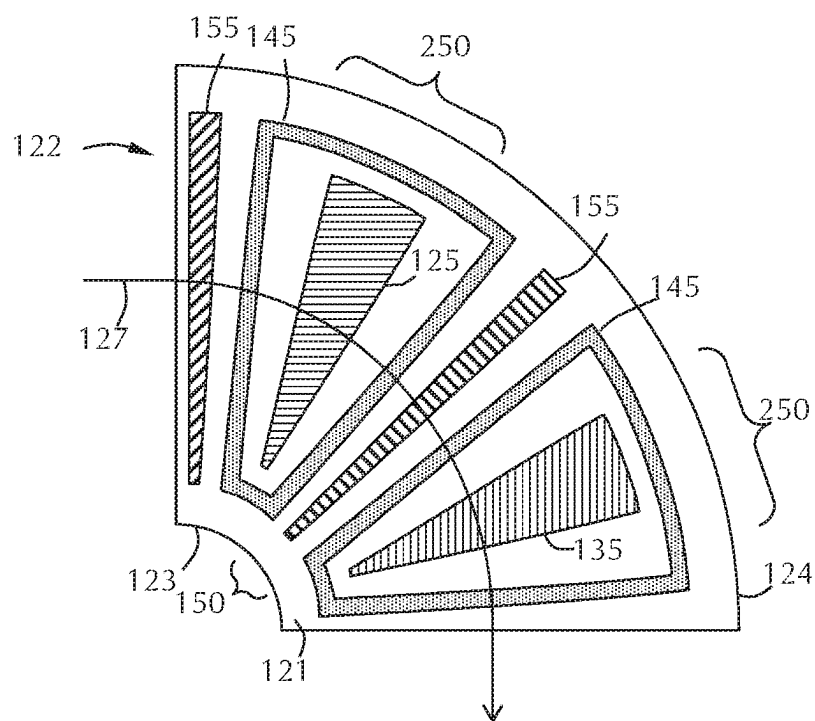
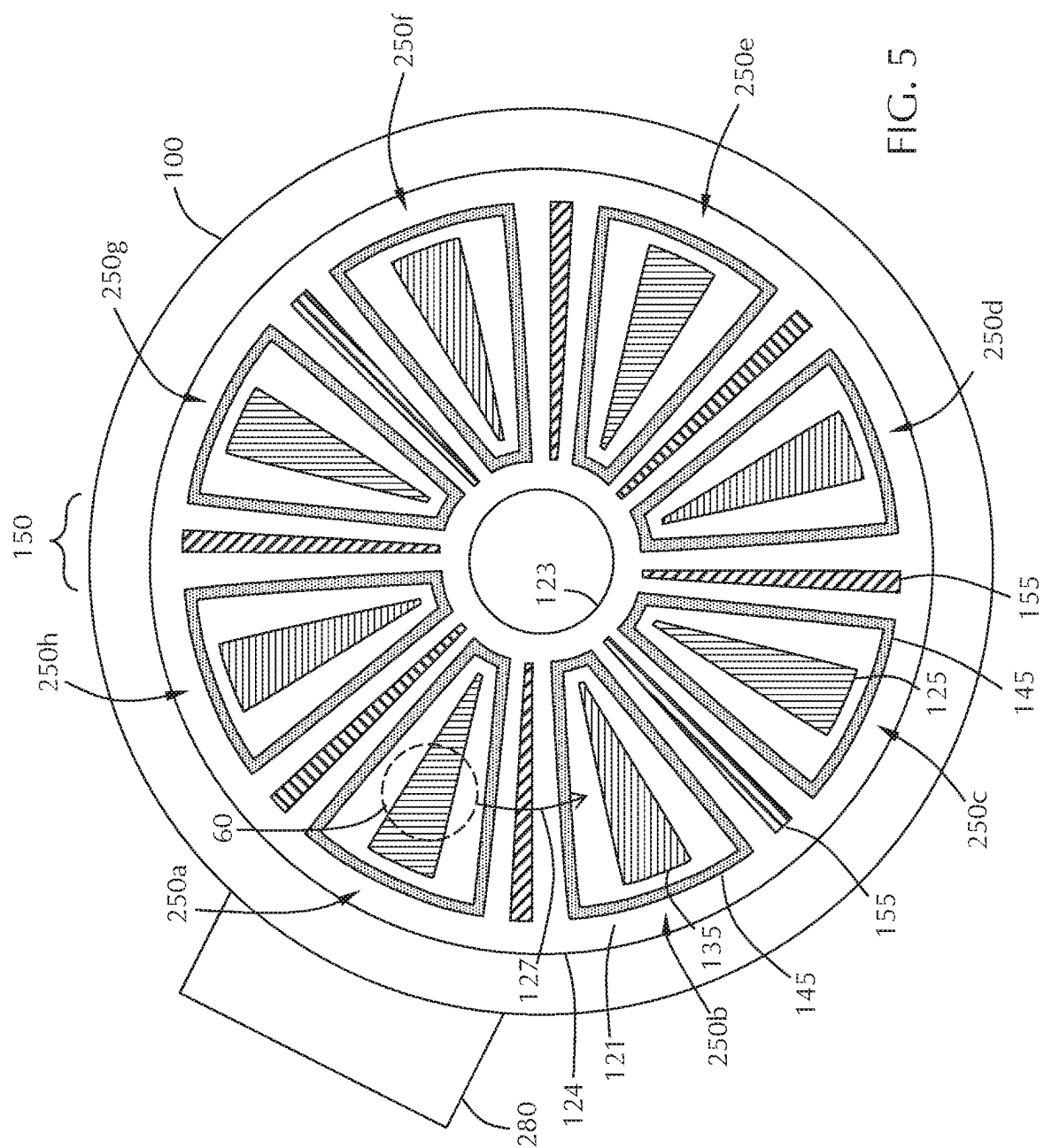


FIG. 4



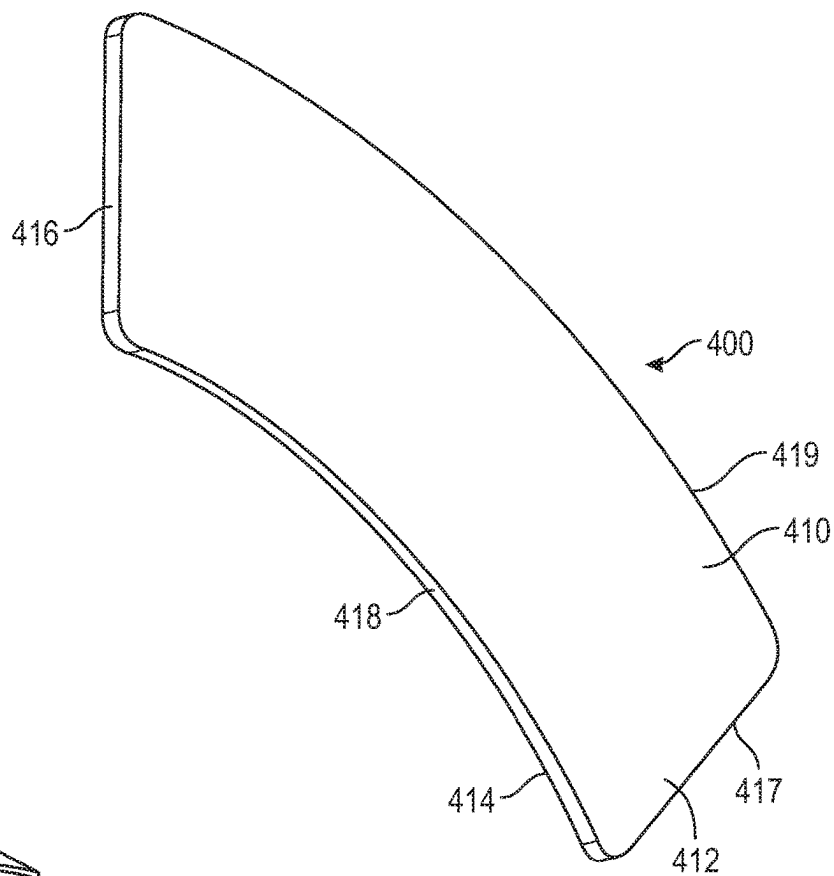


FIG. 6A

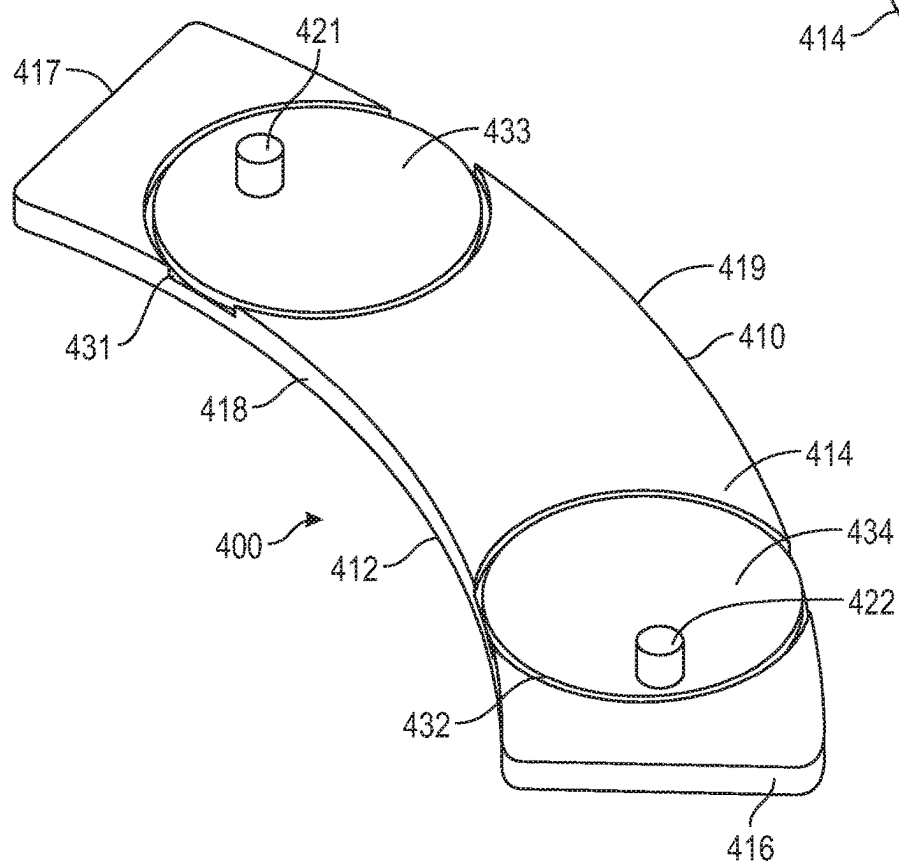


FIG. 6B

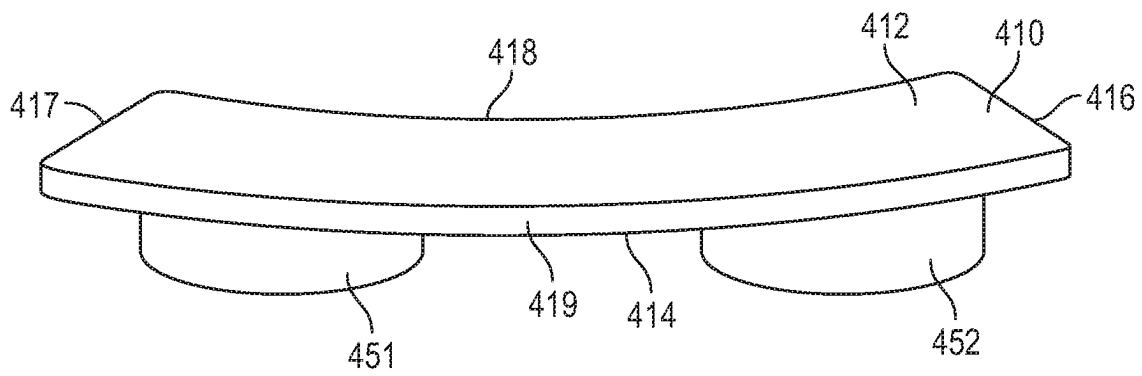


FIG. 7

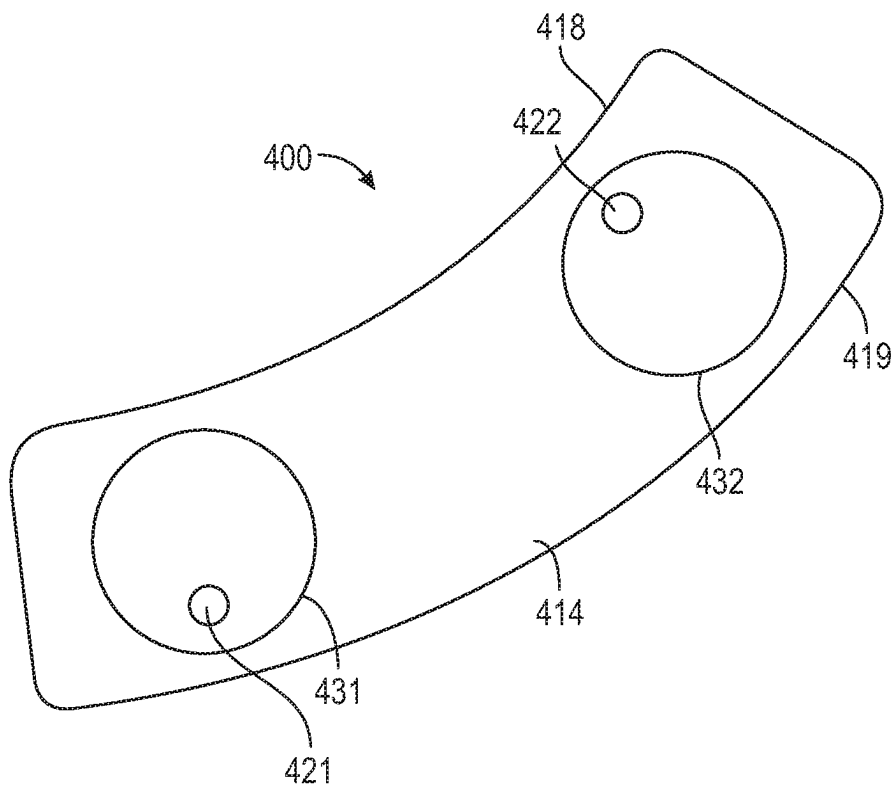


FIG. 8

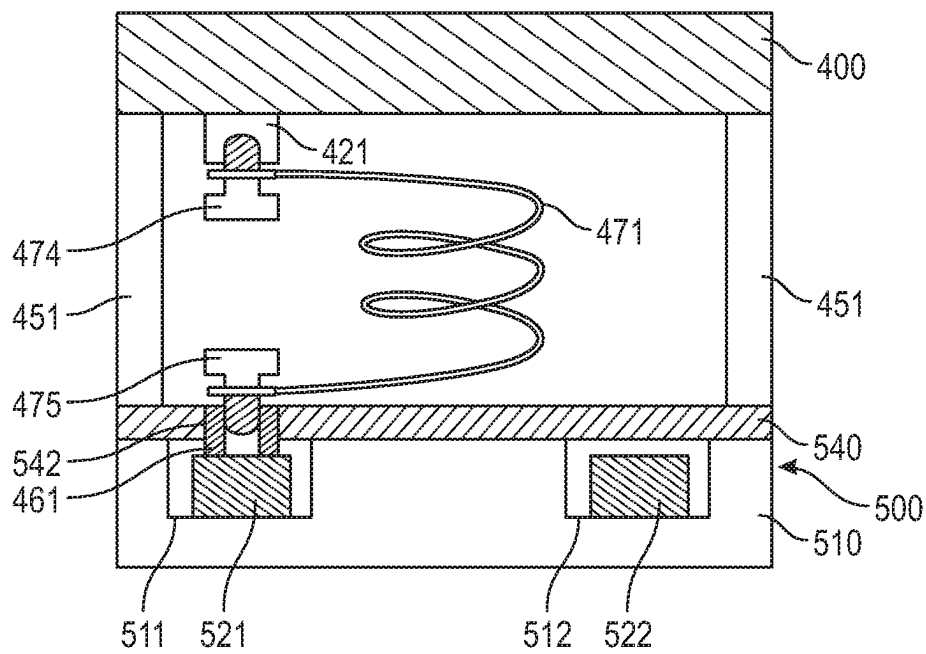


FIG. 9

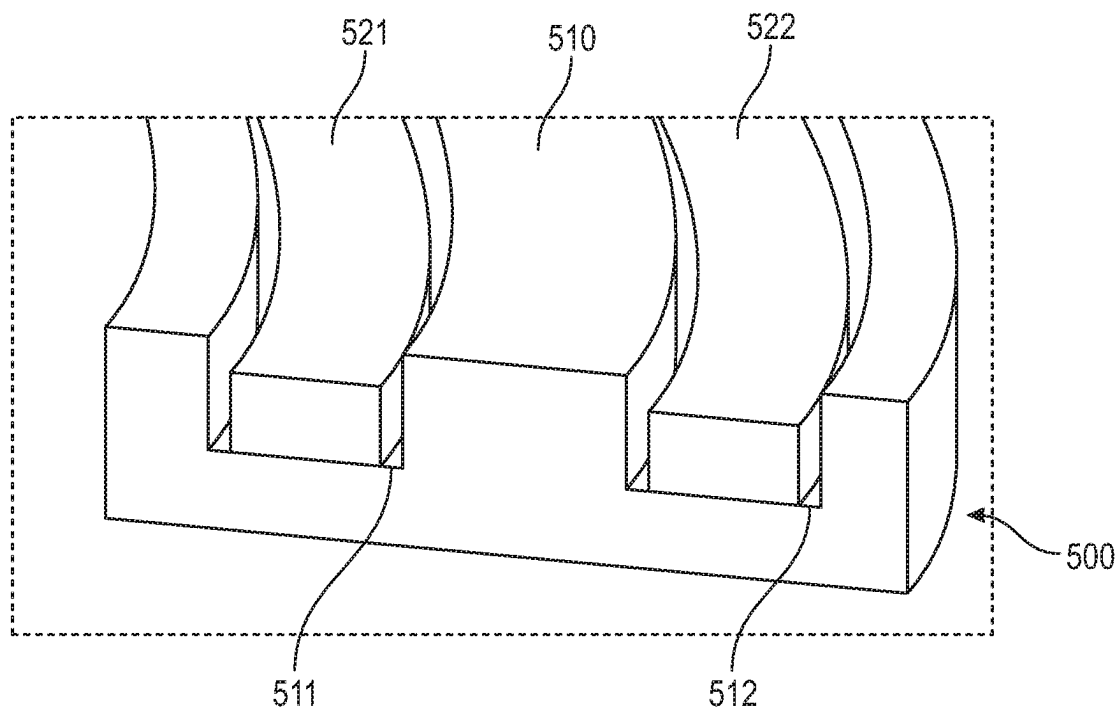


FIG. 10

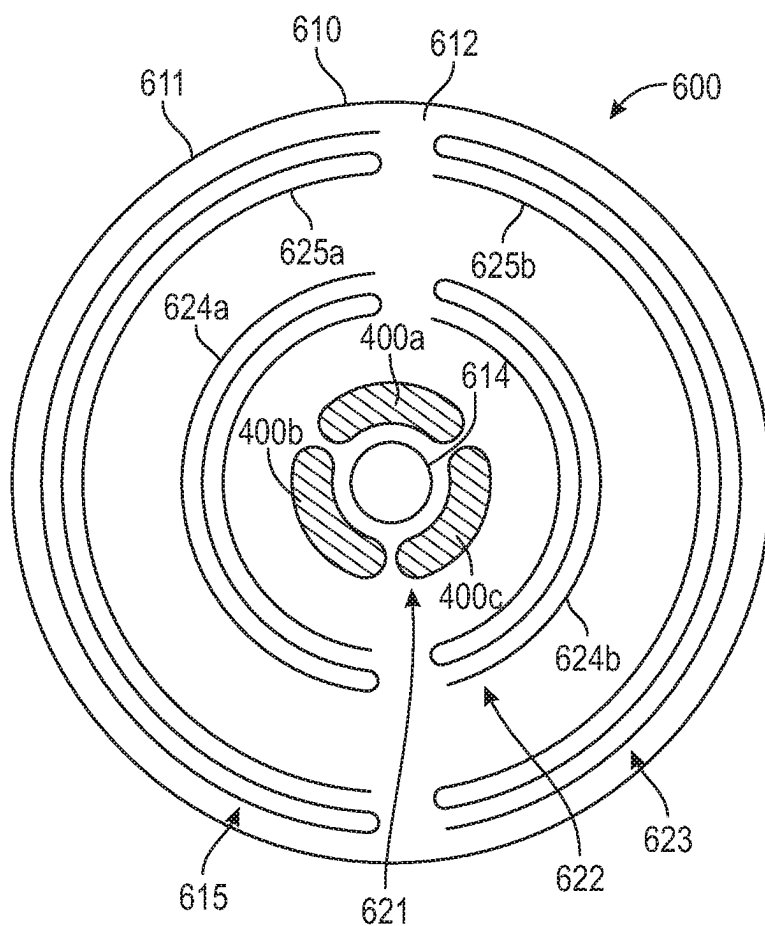


FIG. 11

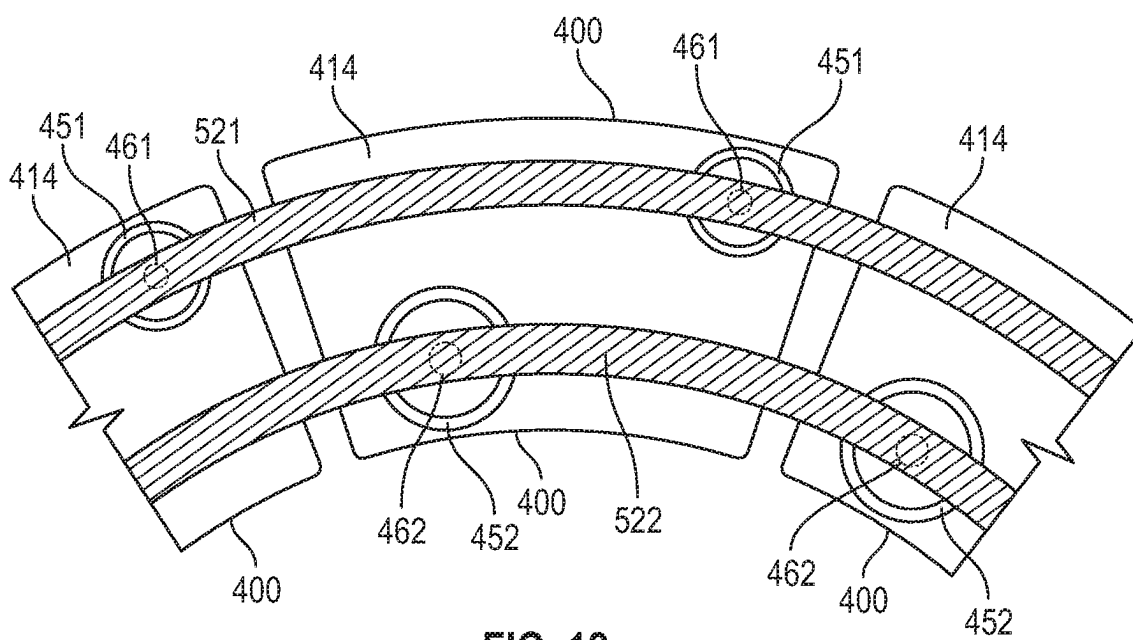


FIG. 12

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PBN HEATERS FOR ALD TEMPERATURE UNIFORMITY

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is the National Stage entry of PCT/US2019/018668, filed on Feb. 20, 2019, which claims priority to U.S. Provisional Application Ser. No. 62/632,748, filed Feb. 20, 2018, the entire disclosures of which are hereby incorporated by reference herein.

FIELD

Embodiments of the disclosure generally relate to apparatus for processing substrates. More particularly, embodiments of the disclosure relate to heaters for batch processing chambers.

BACKGROUND

Atomic Layer Deposition (ALD) and Plasma-Enhanced ALD (PEALD) are deposition techniques that offer control of film thickness and conformality in high-aspect ratio structures. Due to continuously decreasing device dimensions in the semiconductor industry, there is increasing interest and applications that use ALD/PEALD. In some cases, only PEALD can meet specifications for desired film thickness and conformality.

Semiconductor device formation is commonly conducted in substrate processing platforms containing multiple chambers. In some instances, the purpose of a multi-chamber processing platform or cluster tool is to perform two or more processes on a substrate sequentially in a controlled environment. In other instances, however, a multiple chamber processing platform may only perform a single processing step on substrates; the additional chambers are intended to maximize the rate at which substrates are processed by the platform. In the latter case, the process performed on substrates is typically a batch process, wherein a relatively large number of substrates, e.g. 25 or 50, are processed in a given chamber simultaneously. Batch processing is especially beneficial for processes that are too time-consuming to be performed on individual substrates in an economically viable manner, such as for atomic layer deposition (ALD) processes and some chemical vapor deposition (CVD) processes.

During processing, substrates are often heated using tubular heaters which have an upper temperature limit of about 750° C. While the heaters may reach that temperature, the substrate or susceptor assembly being heated typically does not go above about 550° C. The watt density of a tubular heater is high from a central heating wire which radiating 360° from a tubular shape results in a low power density toward the wafer (~30 watts/cm²). Additionally, tubular heaters operating at 750° C. have about a three to six month life span.

Accordingly, there is a need in the art for apparatus that can heat a wafer to temperatures greater than 550° C., have a longer lifetime and/or higher watt densities.

SUMMARY

One or more embodiments of the disclosure are directed to heaters comprising a body having a top and bottom. The body comprises pyrolytic boron nitride (PBN). A first heater

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electrode is connected to the bottom of the body and a second heater electrode is connected to the bottom of the body.

Additional embodiments of the disclosure are directed to heater assemblies comprising a round body having a bottom with an opening in a center of the body and sidewall forming an outer periphery of the body around the bottom. The sidewall and bottom define a cavity within the body. A heater zone is within the cavity of the body. The heater zone comprises one or more heater with a heater body comprising pyrolytic boron nitride (PBN), a first heater electrode connected to a bottom of the heater body and a second heater electrode connected to the bottom of the heater body. A first busbar is electrically connected to the first heater electrode and a second busbar is electrically connected to the second heater electrode and electrically isolated from the first busbar.

BRIEF DESCRIPTION OF THE DRAWINGS

So that the manner in which the above recited features of embodiments of the disclosure can be understood in detail, a more particular description of embodiments of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

FIG. 1 shows a schematic cross-sectional view of a substrate processing system in accordance with one or more embodiments of the disclosure;

FIG. 2 shows a perspective view of a substrate processing system in accordance with one or more embodiment of the disclosure;

FIG. 3 shows a schematic of a substrate processing system in accordance with one or more embodiment of the disclosure;

FIG. 4 shows a schematic view of a front of an injector unit in accordance with one or more embodiment of the disclosure;

FIG. 5 shows a schematic view of a front of a gas distribution assembly in accordance with one or more embodiment of the disclosure;

FIG. 6A shows a top view of a heater in accordance with one or more embodiment of the disclosure;

FIG. 6B shows a bottom view of the heater of FIG. 6A;

FIG. 7 shows a top view of a heater with standoffs in accordance with one or more embodiments of the disclosure;

FIG. 8 shows a bottom view of a heater in accordance with one or more embodiments of the disclosure;

FIG. 9 shows a cross-sectional view of a portion of a heater with standoff and busbars in accordance with one or more embodiments of the disclosure;

FIG. 10 shows a view of a busbar assembly in accordance with one or more embodiment of the disclosure;

FIG. 11 shows a view of a heater assembly in accordance with one or more embodiment of the disclosure; and

FIG. 12 shows a partial view of a heating zone with a plurality of PBN heaters in accordance with one or more embodiment of the disclosure.

DETAILED DESCRIPTION

Embodiments of the disclosure provide a substrate processing system for continuous substrate deposition to maxi-

mize throughput and improve processing efficiency. One or more embodiments of the disclosure are described with respect to a spatial atomic layer deposition chamber.

As used in this specification and the appended claims, the term “substrate” and “wafer” are used interchangeably, both referring to a surface, or portion of a surface, upon which a process acts. It will also be understood by those skilled in the art that reference to a substrate can also refer to only a portion of the substrate, unless the context clearly indicates otherwise. Additionally, reference to depositing on a substrate can mean both a bare substrate and a substrate with one or more films or features deposited or formed thereon.

As used in this specification and the appended claims, the terms “reactive gas”, “precursor”, “reactant”, and the like, are used interchangeably to mean a gas that includes a species which is reactive with a substrate surface. For example, a first “reactive gas” may simply adsorb onto the surface of a substrate and be available for further chemical reaction with a second reactive gas.

As used in this specification and the appended claims, the terms “pie-shaped” and “wedge-shaped” are used interchangeably to describe a body that is a sector of a circle. For example, a wedge-shaped segment may be a fraction of a circle or disc-shaped structure and multiple wedge-shaped segments can be connected to form a circular body. The sector can be defined as a part of a circle enclosed by two radii of a circle and the intersecting arc. The inner edge of the pie-shaped segment can come to a point or can be truncated to a flat edge or rounded. In some embodiments, the sector can be defined as a portion of a ring or annulus.

The path of the substrates can be perpendicular to the gas ports. In some embodiments, each of the gas injector assemblies comprises a plurality of elongate gas ports which extend in a direction substantially perpendicular to the path traversed by a substrate, where a front face of the gas distribution assembly is substantially parallel to the platen. As used in this specification and the appended claims, the term “substantially perpendicular” means that the general direction of movement of the substrates is along a plane approximately perpendicular (e.g., about 45° to 90°) to the axis of the gas ports. For a wedge-shaped gas port, the axis of the gas port can be considered to be a line defined as the mid-point of the width of the port extending along the length of the port.

FIG. 1 shows a cross-section of a processing chamber 100 including a gas distribution assembly 120, also referred to as injectors or an injector assembly, and a susceptor assembly 140. The gas distribution assembly 120 is any type of gas delivery device used in a processing chamber. The gas distribution assembly 120 includes a front surface 121 which faces the susceptor assembly 140. The front surface 121 can have any number or variety of openings to deliver a flow of gases toward the susceptor assembly 140. The gas distribution assembly 120 also includes an outer peripheral edge 124 which in the embodiments shown, is substantially round.

The specific type of gas distribution assembly 120 used can vary depending on the particular process being used. Embodiments of the disclosure can be used with any type of processing system where the gap between the susceptor and the gas distribution assembly is controlled. While various types of gas distribution assemblies can be employed (e.g., showerheads), embodiments of the disclosure may be particularly useful with spatial ALD gas distribution assemblies which have a plurality of substantially parallel gas channels. As used in this specification and the appended claims, the term “substantially parallel” means that the elongate axis of the gas channels extend in the same general direction. There

can be slight imperfections in the parallelism of the gas channels. The plurality of substantially parallel gas channels can include at least one first reactive gas A channel, at least one second reactive gas B channel, at least one purge gas P channel and/or at least one vacuum V channel. The gases flowing from the first reactive gas A channel(s), the second reactive gas B channel(s) and the purge gas P channel(s) are directed toward the top surface of the wafer. Some of the gas flow moves horizontally across the surface of the wafer and out of the processing region through the purge gas P channel(s). A substrate moving from one end of the gas distribution assembly to the other end will be exposed to each of the process gases in turn, forming a layer on the substrate surface.

In some embodiments, the gas distribution assembly 120 is a rigid stationary body made of a single injector unit. In one or more embodiments, the gas distribution assembly 120 is made up of a plurality of individual sectors (e.g., injector units 122), as shown in FIG. 2. Either a single piece body or a multi-sector body can be used with the various embodiments of the disclosure described.

The susceptor assembly 140 is positioned beneath the gas distribution assembly 120. The susceptor assembly 140 includes a top surface 141 and at least one recess 142 in the top surface 141. The susceptor assembly 140 also has a bottom surface 143 and an edge 144. The recess 142 can be any suitable shape and size depending on the shape and size of the substrates 60 being processed. In the embodiment shown in FIG. 1, the recess 142 has a flat bottom to support the bottom of the wafer; however, the bottom of the recess can vary. In some embodiments, the recess has step regions around the outer peripheral edge of the recess which are sized to support the outer peripheral edge of the wafer. The amount of the outer peripheral edge of the wafer that is supported by the steps can vary depending on, for example, the thickness of the wafer and the presence of features already present on the back side of the wafer.

In some embodiments, as shown in FIG. 1, the recess 142 in the top surface 141 of the susceptor assembly 140 is sized so that a substrate 60 supported in the recess 142 has a top surface 61 substantially coplanar with the top surface 141 of the susceptor 140. As used in this specification and the appended claims, the term “substantially coplanar” means that the top surface of the wafer and the top surface of the susceptor assembly are coplanar within ± 0.2 mm. In some embodiments, the top surfaces are coplanar within ± 0.15 mm, ± 0.10 mm or ± 0.05 mm. The recess 142 of some embodiments supports a wafer so that the inner diameter (ID) of the wafer is located within the range of about 170 mm to about 185 mm from the center (axis of rotation) of the susceptor. In some embodiments, the recess 142 supports a wafer so that the outer diameter (OD) of the wafer is located in the range of about 470 mm to about 485 mm from the center (axis of rotation) of the susceptor.

The susceptor assembly 140 of FIG. 1 includes a support post 160 which is capable of lifting, lowering and rotating the susceptor assembly 140. The susceptor assembly may include a heater, or gas lines, or electrical components within the center of the support post 160. The support post 160 may be the primary means of increasing or decreasing the gap between the susceptor assembly 140 and the gas distribution assembly 120, moving the susceptor assembly 140 into proper position. The susceptor assembly 140 may also include fine tuning actuators 162 which can make micro-adjustments to susceptor assembly 140 to create a predetermined gap 170 between the susceptor assembly 140 and the gas distribution assembly 120. In some embodi-

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ments, the gap **170** distance is in the range of about 0.1 mm to about 5.0 mm, or in the range of about 0.1 mm to about 3.0 mm, or in the range of about 0.1 mm to about 2.0 mm, or in the range of about 0.2 mm to about 1.8 mm, or in the range of about 0.3 mm to about 1.7 mm, or in the range of about 0.4 mm to about 1.6 mm, or in the range of about 0.5 mm to about 1.5 mm, or in the range of about 0.6 mm to about 1.4 mm, or in the range of about 0.7 mm to about 1.3 mm, or in the range of about 0.8 mm to about 1.2 mm, or in the range of about 0.9 mm to about 1.1 mm, or about 1 mm.

The processing chamber **100** shown in the Figures is a carousel-type chamber in which the susceptor assembly **140** can hold a plurality of substrates **60**. As shown in FIG. 2, the gas distribution assembly **120** may include a plurality of separate injector units **122**, each injector unit **122** being capable of depositing a film on the wafer, as the wafer is moved beneath the injector unit. Two pie-shaped injector units **122** are shown positioned on approximately opposite sides of and above the susceptor assembly **140**. This number of injector units **122** is shown for illustrative purposes only. It will be understood that more or less injector units **122** can be included. In some embodiments, there are a sufficient number of pie-shaped injector units **122** to form a shape conforming to the shape of the susceptor assembly **140**. In some embodiments, each of the individual pie-shaped injector units **122** may be independently moved, removed and/or replaced without affecting any of the other injector units **122**. For example, one segment may be raised to permit a robot to access the region between the susceptor assembly **140** and gas distribution assembly **120** to load/unload substrates **60**.

Processing chambers having multiple gas injectors can be used to process multiple wafers simultaneously so that the wafers experience the same process flow. For example, as shown in FIG. 3, the processing chamber **100** has four gas injector assemblies and four substrates **60**. At the outset of processing, the substrates **60** can be positioned between the injector assemblies **30**. Rotating **17** the susceptor assembly **140** by 45° will result in each substrate **60** which is between gas distribution assemblies **120** to be moved to an gas distribution assembly **120** for film deposition, as illustrated by the dotted circle under the gas distribution assemblies **120**. An additional 45° rotation would move the substrates **60** away from the injector assemblies **30**. With spatial ALD injectors, a film is deposited on the wafer during movement of the wafer relative to the injector assembly. In some embodiments, the susceptor assembly **140** is rotated in increments that prevent the substrates **60** from stopping beneath the gas distribution assemblies **120**. The number of substrates **60** and gas distribution assemblies **120** can be the same or different. In some embodiments, there is the same number of wafers being processed as there are gas distribution assemblies. In one or more embodiments, the number of wafers being processed are fraction of or an integer multiple of the number of gas distribution assemblies. For example, if there are four gas distribution assemblies, there are 4x wafers being processed, where x is an integer value greater than or equal to one.

The processing chamber **100** shown in FIG. 3 is merely representative of one possible configuration and should not be taken as limiting the scope of the disclosure. Here, the processing chamber **100** includes a plurality of gas distribution assemblies **120**. In the embodiment shown, there are four gas distribution assemblies (also called injector assemblies **30**) evenly spaced about the processing chamber **100**. The processing chamber **100** shown is octagonal, however, those skilled in the art will understand that this is one

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possible shape and should not be taken as limiting the scope of the disclosure. The gas distribution assemblies **120** shown are trapezoidal, but can be a single circular component or made up of a plurality of pie-shaped segments, like that shown in FIG. 2.

The embodiment shown in FIG. 3 includes a load lock chamber **180**, or an auxiliary chamber like a buffer station. This chamber **180** is connected to a side of the processing chamber **100** to allow, for example the substrates (also referred to as substrates **60**) to be loaded/unloaded from the processing chamber **100**. A wafer robot may be positioned in the chamber **180** to move the substrate onto the susceptor.

Rotation of the carousel (e.g., the susceptor assembly **140**) can be continuous or discontinuous. In continuous processing, the wafers are constantly rotating so that they are exposed to each of the injectors in turn. In discontinuous processing, the wafers can be moved to the injector region and stopped, and then to the region **84** between the injectors and stopped. For example, the carousel can rotate so that the wafers move from an inter-injector region across the injector (or stop adjacent the injector) and on to the next inter-injector region where the carousel can pause again. Pausing between the injectors may provide time for additional processing steps between each layer deposition (e.g., exposure to plasma).

FIG. 4 shows a sector or portion of a gas distribution assembly **220**, which may be referred to as an injector unit **122**. The injector units **122** can be used individually or in combination with other injector units. For example, as shown in FIG. 5, four of the injector units **122** of FIG. 4 are combined to form a single gas distribution assembly **220**. (The lines separating the four injector units are not shown for clarity.) While the injector unit **122** of FIG. 4 has both a first reactive gas port **125** and a second reactive gas port **135** in addition to purge gas ports **155** and vacuum ports **145**, an injector unit **122** does not need all of these components.

Referring to both FIGS. 4 and 5, a gas distribution assembly **220** in accordance with one or more embodiment may comprise a plurality of sectors (or injector units **122**) with each sector being identical or different. The gas distribution assembly **220** is positioned within the processing chamber and comprises a plurality of elongate gas ports **125**, **135**, **145** in a front surface **121** of the gas distribution assembly **220**. The plurality of elongate gas ports **125**, **135**, **145** and vacuum ports **155** extend from an area adjacent the inner peripheral edge **123** toward an area adjacent the outer peripheral edge **124** of the gas distribution assembly **220**. The plurality of gas ports shown include a first reactive gas port **125**, a second reactive gas port **135**, a vacuum port **145** which surrounds each of the first reactive gas ports and the second reactive gas ports and a purge gas port **155**.

With reference to the embodiments shown in FIG. 4 or 5, when stating that the ports extend from at least about an inner peripheral region to at least about an outer peripheral region, however, the ports can extend more than just radially from inner to outer regions. The ports can extend tangentially as vacuum port **145** surrounds reactive gas port **125** and reactive gas port **135**. In the embodiment shown in FIGS. 4 and 5, the wedge shaped reactive gas ports **125**, **135** are surrounded on all edges, including adjacent the inner peripheral region and outer peripheral region, by a vacuum port **145**.

Referring to FIG. 4, as a substrate moves along path **127**, each portion of the substrate surface is exposed to the various reactive gases. To follow the path **127**, the substrate will be exposed to, or "see", a purge gas port **155**, a vacuum port **145**, a first reactive gas port **125**, a vacuum port **145**, a

purge gas port **155**, a vacuum port **145**, a second reactive gas port **135** and a vacuum port **145**. Thus, at the end of the path **127** shown in FIG. **4**, the substrate has been exposed to gas streams from the first reactive gas port **125** and the second reactive gas port **135** to form a layer. The injector unit **122** shown makes a quarter circle but could be larger or smaller. The gas distribution assembly **220** shown in FIG. **5** can be considered a combination of four of the injector units **122** of FIG. **4** connected in series.

The injector unit **122** of FIG. **4** shows a gas curtain **150** that separates the reactive gases. The term "gas curtain" is used to describe any combination of gas flows or vacuum that separate reactive gases from mixing. The gas curtain **150** shown in FIG. **4** comprises the portion of the vacuum port **145** next to the first reactive gas port **125**, the purge gas port **155** in the middle and a portion of the vacuum port **145** next to the second reactive gas port **135**. This combination of gas flow and vacuum can be used to prevent or minimize gas phase reactions of the first reactive gas and the second reactive gas.

Referring to FIG. **5**, the combination of gas flows and vacuum from the gas distribution assembly **220** form a separation into a plurality of processing regions **250**. The processing regions are roughly defined around the individual reactive gas ports **125**, **135** with the gas curtain **150** between **250**. The embodiment shown in FIG. **5** makes up eight separate processing regions **250** with eight separate gas curtains **150** between. A processing chamber can have at least two processing region. In some embodiments, there are at least three, four, five, six, seven, eight, nine, 10, 11 or 12 processing regions.

During processing a substrate may be exposed to more than one processing region **250** at any given time. However, the portions that are exposed to the different processing regions will have a gas curtain separating the two. For example, if the leading edge of a substrate enters a processing region including the second reactive gas port **135**, a middle portion of the substrate will be under a gas curtain **150** and the trailing edge of the substrate will be in a processing region including the first reactive gas port **125**.

A factory interface **280**, which can be, for example, a load lock chamber, is shown connected to the processing chamber **100**. A substrate **60** is shown superimposed over the gas distribution assembly **220** to provide a frame of reference. The substrate **60** may often sit on a susceptor assembly to be held near the front surface **121** of the gas distribution assembly **120** (also referred to as a gas distribution plate). The substrate **60** is loaded via the factory interface **280** into the processing chamber **100** onto a substrate support or susceptor assembly (see FIG. **3**). The substrate **60** can be shown positioned within a processing region because the substrate is located adjacent the first reactive gas port **125** and between two gas curtains **150a**, **150b**. Rotating the substrate **60** along path **127** will move the substrate counter-clockwise around the processing chamber **100**. Thus, the substrate **60** will be exposed to the first processing region **250a** through the eighth processing region **250h**, including all processing regions between. For each cycle around the processing chamber, using the gas distribution assembly shown, the substrate **60** will be exposed to four ALD cycles of first reactive gas and second reactive gas.

The conventional ALD sequence in a batch processor, like that of FIG. **5**, maintains chemical A and B flow respectively from spatially separated injectors with pump/purge section between. The conventional ALD sequence has a starting and ending pattern which might result in non-uniformity of the deposited film. The inventors have surprisingly discovered

that a time based ALD process performed in a spatial ALD batch processing chamber provides a film with higher uniformity. The basic process of exposure to gas A, no reactive gas, gas B, no reactive gas would be to sweep the substrate under the injectors to saturate the surface with chemical A and B respectively to avoid having a starting and ending pattern form in the film. The inventors have surprisingly found that the time based approach is especially beneficial when the target film thickness is thin (e.g., less than 20 ALD cycles), where starting and ending pattern have a significant impact on the within wafer uniformity performance. The inventors have also discovered that the reaction process to create SiCN, SiCO and SiCON films, as described herein, could not be accomplished with a time-domain process. The amount of time used to purge the processing chamber results in the stripping of material from the substrate surface. The stripping does not happen with the spatial ALD process described because the time under the gas curtain is short.

Accordingly, embodiments of the disclosure are directed to processing methods comprising a processing chamber **100** with a plurality of processing regions **250a-250h** with each processing region separated from an adjacent region by a gas curtain **150**. For example, the processing chamber shown in FIG. **5**. The number of gas curtains and processing regions within the processing chamber can be any suitable number depending on the arrangement of gas flows. The embodiment shown in FIG. **5** has eight gas curtains **150** and eight processing regions **250a-250h**. The number of gas curtains is generally equal to or greater than the number of processing regions. For example, if region **250a** had no reactive gas flow, but merely served as a loading area, the processing chamber would have seven processing regions and eight gas curtains.

A plurality of substrates **60** are positioned on a substrate support, for example, the susceptor assembly **140** shown FIGS. **1** and **2**. The plurality of substrates **60** are rotated around the processing regions for processing. Generally, the gas curtains **150** are engaged (gas flowing and vacuum on) throughout processing including periods when no reactive gas is flowing into the chamber.

A first reactive gas A is flowed into one or more of the processing regions **250** while an inert gas is flowed into any processing region **250** which does not have a first reactive gas A flowing into it. For example if the first reactive gas is flowing into processing regions **250b** through processing region **250h**, an inert gas would be flowing into processing region **250a**. The inert gas can be flowed through the first reactive gas port **125** or the second reactive gas port **135**.

Referring again to FIG. **1**, some embodiments of the disclosure incorporate a heater **300** located adjacent the bottom surface **143** of the susceptor assembly **140**. The heater **300** can be spaced from the bottom surface **143** by any suitable distance or can be in direct contact with the bottom surface **143**. The heater **300** illustrated is a disc-shaped component with a central opening **305** through which the support post **160** extends. The heater **300** can be connected to the support post **160** so that the heater **300** moves with the susceptor assembly **140** so that the distance from the bottom surface **143** remains the same. In some embodiments, the heater **300** rotates with the susceptor assembly **140**. In some embodiments, the heater **300** is independent from the susceptor assembly **140** in that the movement of the heater **300** is separate from and independently controlled than the susceptor assembly **140**.

The heater **300** illustrated in FIG. **1** includes heating elements **310**. Each of the heating elements **310** can be separate elements independently controlled or can be a

uniform coil of material that extends around the opening **305** forming a spiral shape when viewed from above. The heating elements **310** illustrated are arranged in three radial zones so that each zone is located at a different distance from the central opening **305**. The inner zone **315a** is closest zone to the support post **160** at the center of the susceptor assembly **140**. The inner zone **315a** is illustrated as three coils of heating elements **310a** which can be a single coil or multiple coils. In some embodiments, the heating elements in any of the zones are separated into rotational zones. For example, in the illustrated embodiment, the left side of the heater **300** can have different coils than the right side, so that each of the radial zones has two rotational zones.

The second zone **315b** is illustrated as being located below the recess **142** that supports the substrate **60**. The heating elements **310b** in the second zone **315b** are shown closer to the bottom surface **143** of the susceptor assembly **140** than the heating elements **310a** of the inner zone **315a**. In some embodiments, the heating elements **310a** of the inner zone **315a** are closer to the bottom surface **143** than the second zone **315b** heating elements **310b**. In some embodiments, the heating elements **310a** of the inner zone **315a** and the heating elements **310b** of the second zone **315b** are about the same distance from the bottom surface **143**.

The heating elements **310a** of the first zone **315a** are separated from the heating elements **310b** of the second zone **315b** by a first shield **320a**. The size and shape of the first shield **320a** can be any suitable dimensions and can be positioned at any distance from the bottom surface **143** of the susceptor assembly **140**. In some embodiments, there is no first shield **320a** separating the inner zone **315a** from the second zone **315b**.

The heating elements **310c** of the outer zone **315c** are illustrated as being located at the outer portion of the susceptor assembly **140**. In some embodiments, the heating elements **310c** of the outer zone **315c** are separated from the heating elements **310b** of the second zone **315b** by a second shield **320b**. In some embodiments, the heating elements **310c** of the outer zone **315c** are a different distance from the bottom surface **143** of the susceptor assembly **140** than one or more of the inner zone **315a** and/or the second zone **315b**. In some embodiments, the heater **300** includes more or less than three zones. For example, in some embodiments, there are four heater zones (not shown), an inner heating zone, a second heating zone, a third heating zone and an outer heating zone.

One or more embodiments of the disclosure advantageously provide heaters that can heat a wafer to 800° C. or higher. Some embodiments advantageously provide pyrolytic boron nitride/pyrolytic graphite (PBN/PG) heaters that can safely reach surface temperature of about 1200° C. Some embodiments of the disclosure provide apparatus that can provide wafer temperature uniformity less than or equal to about 2° C. Some embodiments provide heater that have very high watt densities (up to 100 watt/cm²) from large flat surfaces.

The batch processing chamber of some embodiments uses a large diameter graphite susceptor (plate) to support, heat and allow processing of six wafers simultaneously. The plate rotates during processing and receives heat from the heaters fixed in a chamber cavity below. The cavity is below the susceptor and is formed by a fluid cooled body maintained at a low temperature (e.g., 40-60° C.). The cavity provides penetrations to feed electrical power for a number of zones of heating, pump sensing, susceptor positioning camera viewing and human eye viewing. The heating zones can be

positioned in the cavity at any elevation (e.g., 35 mm to 150 mm) below the susceptor as the susceptor can translate downward to transfer wafers.

PBN/PG heaters can provide a flat surface with extremely high watt densities (e.g., up to 100 W/cm²). Multiple PBN heaters may be connected to a common zone power supply to a pair of electrically isolated busbars inside the chamber allowing an array of PBN elements per zone. The PBN elements, in parallel connection to the common power supply, may have equal resistances to provide equal power outputs and operating temperatures.

Some embodiments advantageously provide flat plate PBN heaters with high density upward directed energy at the graphite susceptor in radially discrete control zones. For example, three separate control zones can provide the ability to manage uniform wafer temperature to less than one degree.

In some embodiments, the inner zone heater is replaced with a PBN heater. The inner zone PBN heater may provide good temperature uniformity by introducing high powers at the center of the susceptor. The replaced inner zone tubular heater may not provide sufficient power at the center of the susceptor due to, for example, high thermal losses down the rotation shaft and up to the aluminum injector. The PBN heater of some embodiments has a smaller outside diameter than a tubular heater with a higher watt density and may focus the energy nearer the center of the susceptor. In some embodiments, the flat PBN heaters can be located closer to the susceptor to provide more efficient temperature control than can be achieved with a tubular heater.

Referring to FIGS. 6A, 6B and 7, one or more embodiments of the disclosure are directed to heaters **400**. FIG. 6A shows a top view of the heater **400** and FIG. 6B shows a bottom view of the heater **400**. As used in this manner, the relative terms “top” and “bottom” are used to describe different views of the heater **400** and should not be taken as implying a specific spatial direction. The heater **400** has a body **410** with a top **412** and a bottom **414**.

In some embodiments, the body **410** is a rectangular shaped component with straight sides. In some embodiments, as illustrated, the body **410** is a curved component having a first end **416** and a second end **417** connected by an arc-shaped inner end **418** and an arc-shaped outer end **419**.

The body **410** can be made of any suitable material. In some embodiments, the body **410** comprises pyrolytic boron nitride (PBN), pyrolytic graphite (PG) or a mixture of PBN/PG. In some embodiments, the mixture of PBN/PG has a ratio of PBN:PG in the range of about 100:1 to about 1:100. In some embodiments, the body **410** of the heater **400** consists essentially of PBN. As used in this manner, the term “consists essentially of PBN” means that the composition is greater than 99% or 99.5% PBN, on a weight basis.

The heater **400** includes a first heater electrode **421** and a second heater electrode **422** connected to the bottom **414** of the body **410**. The first heater electrode **421** and the second heater electrode **422** can be made of any suitable material that can efficiently conduct electricity. In some embodiments, the first heater electrode **421** and/or the second heater electrode **422** are made of material comprising molybdenum.

The positioning of the heater electrodes can vary depending on, for example, the location of power source connections. In some embodiments, the first heater electrode **421** is located closer to the inner end **418** or the outer end **419** than the second heater electrode **422**. FIG. 8 illustrates a bottom view of the heater **400** showing the first heater electrode **421**

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closer to the outer end **419** than the inner end **418** and the second heater electrode **422** closer to the inner end **418** than to the outer end **419**.

Some embodiments of the heater **400** include a first recess **431** and a second recess **432** on the bottom **414** of the body **410**. The recesses can be any suitable shape and width. In the embodiment illustrated in FIG. 6B, the recesses **431**, **432** are circular and the first heater electrode **421** is within the bounds of the first recess **431** and the second heater electrode **422** is within the bounds of the second recess **432**. In the illustrated embodiments, there is a first raised portion **433** in the first recess **431** and a second raised portion **434** in the second recess **432**. The raised portions **433**, **434** are sized so that the recesses **431**, **432** have a suitable width to support a separate component. In some embodiments, the recesses **431**, **432** have an outer diameter in the range of about 40 mm to about 150 mm, or in the range of about 50 mm to about 140 mm, or in the range of about 60 mm to about 130 mm in some embodiments, the width of the recesses **431**, **432** are greater than or equal to about 2 mm, 3 mm, 4 mm, 5 mm, 10 mm or 15 mm.

In some embodiments, as illustrated in FIG. 7, a first standoff **451** is positioned in the first recess **431** and a second standoff **452** is positioned in the second recess **432**. The standoffs **451**, **452** can be made of any suitable electrically insulating (i.e., non-conductive) material that can be used to isolate power connections. In some embodiments, the standoffs **451**, **452** are quartz. In some embodiments, the standoffs have a bottom comprising an electrically insulating material, for example, quartz.

The connection to the power source can occur through the heater electrodes **421**, **422** by connection with a busbar connector. The busbar connector can be formed into the bottom of and extending through the standoffs or can be a separate component connected to the electrodes. The busbar connectors can be considered a part of the standoffs even when a separate component. In some embodiments, as shown in FIG. 9, the first busbar connector **461** is a separate component that passes through a busbar cover **540** to contact the first busbar **521**.

In some embodiments, a first wire **471** connects the first busbar connector **461** to the first heater electrode **421**. In a similar arrangement, a second wire (not shown) connects the second busbar connector (not shown) to the second heater electrode **422**. The first wire **471** is located within the first standoff **451** and the second wire is within the second standoff.

The first wire **471** can be connected to the first heater electrode **421** and the first busbar connector **461** by any suitable connector. In some embodiments, a screw **474** and, optionally, washer (not shown) connect the first wire **471** to the first heater electrode **421**. In some embodiments, a screw **475** and, optionally, washer or connecting ring (not numbered), connect the first wire **471** to the first busbar connector **461**. The second wire can be connected to the second heater electrode and the second busbar connector by a suitable connector similar to that of the first wire **471**. In some embodiments, the wire comprises braided molybdenum. In some embodiments, the screw and optional washer are molybdenum.

Referring to FIG. 10, some embodiments include a busbar assembly **500**. The busbar assembly **500** can include an electrically insulating busbar housing **510** (also called a raceway) comprising two channels **511**, **512**. A first busbar **521** can be positioned within the first channel **511** and a second busbar **522** can be positioned within the second channel **512**. In use, the first busbar **521** is in electrical

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communication with the first busbar connector **461** and the second busbar **522** is in electrical communication with the second busbar connector **462**. The first busbar **521** and the second busbar **522** can be connected to one or more power source to provide a voltage differential between the first busbar **521** and the second busbar **522**.

The busbar housing **510** can be made from any suitable electrically insulating material. In some embodiments, the busbar housing **510** comprises alumina. In some embodiments, the busbar housing **510** is made of segmented alumina which may act to conduct heat away from the busbars during use.

Referring back to FIG. 9, some embodiments of the busbar assembly **500** include a busbar cover **540**. The busbar cover **540** can have a channel or opening **542** to allow the busbar connector **461**, **462** to make contact with the busbar **521**, **522**.

Referring to FIG. 11, some embodiments of the disclosure are directed to heater assemblies **600**. The heater assembly **600** of some embodiments has a round body **610** with a bottom **612** and an opening **614** in the center of the body **610**. A sidewall **611** forms an outer periphery of the body **610** around the bottom **612**. The sidewall **611** and bottom **612** define a cavity **615** within the body **610**.

The heater assembly **600** includes at least one heater zone **621**. In the embodiment illustrated in FIG. 11, there are three radial zones: an inner zone **621**, a second zone **622** and an outer zone **623**. Each of the radial zones can be controlled independently of the other zones. In some embodiments, there are more than three radial zones. Any of the zones can be separated from an adjacent zone by a heat shield (as shown in FIG. 1). The inner zone is the zone closest to the opening **614** and may be referred to as the first zone. The outer zone is the zone closest to the sidewall **611**. The zones between the inner zone **621** and the outer zone **623** are referred to as the second zone, third zone, fourth zone, etc. The inner zone may be located a distance from the opening **614** that allows the heating zone to be formed around the periphery of the support post **160** (see FIG. 1).

Each radial zone can be made up of one or more rotational zones. In the illustrated embodiment, the inner zone **621** has three rotational zones made up of heater **400a**, heater **400b** and heater **400c**. The second zone **622** has two rotational zones made up of heating element **624a** and heating element **624b**. The outer zone **623** also has two rotation zones made up of heating element **625a** and heating element **625b**. In some embodiments, each of the radial zones has the same number of rotational zones.

In the illustrated embodiment, the inner zone **621** comprises PBN heaters **400** and the second zone **622** and outer zone **623** are tubular heating elements. In some embodiments, the inner zone **621**, second zone **622** and outer zone **623** comprise PBN heaters **400**.

FIG. 12 shows a portion of a heating zone with a plurality of heaters **400**. The busbars **521**, **522** are shown not in a busbar housing for clarity purposes. The first standoff **451** and first busbar connector **461** are located to contact the first busbar **521**. The second standoff **452** and second busbar connector **462** are located to contact the second busbar **522**. The plurality of heaters **400** can be arranged to form a heating zone that encircles the opening **614**.

Some embodiments of the disclosure are directed to processing chambers incorporating the heater **400** or heating assembly **600**. The heating assembly **600** is positioned below the susceptor assembly **140** around the support post **160**. One or more heat shields can be positioned between the different heating zones.

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While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

What is claimed is:

1. A heater comprising:
 - a body comprising pyrolytic boron nitride (PBN), the body having a top and a bottom, the body having a first end and a second end connected by an arc-shaped inner end and an arc-shaped outer end, the bottom of the body having a first recess and a second recess, each of the first recess and the second recess having a raised portion therein, the raised portion in each of the first recess and the second recess extending toward the bottom of the body;
 - a first heater electrode connected to the bottom of the body, the first heater electrode within the first recess;
 - a second heater electrode connected to the bottom of the body, the second heater electrode within the second recess; and
 - a first standoff positioned within the first recess and enclosing the first heater electrode, and a second standoff positioned within the second recess and enclosing the second heater electrode, the first standoff and the second standoff comprising an electrically insulating material.
2. The heater of claim 1, wherein the body of the heater consists of PBN.
3. The heater of claim 1, wherein the first standoff and the second standoff comprise quartz.
4. The heater of claim 1, wherein the first standoff and the second standoff have a bottom comprising an electrically insulating material.
5. The heater of claim 4, wherein the first standoff has a first busbar connector extending through the bottom of the first standoff and the second standoff has a second busbar connector extending through the bottom of the second standoff.
6. The heater of claim 5, further comprising a first wire connecting the first busbar connector to the first heater electrode and a second wire connecting the second busbar

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connector to the second heater electrode, the first wire within the first standoff and the second wire within the second standoff.

7. The heater of claim 6, further comprising a first busbar and a second busbar, the first busbar in electrical communication with the first busbar connector and the second busbar in electrical communication with the second busbar connector.

8. The heater of claim 7, wherein the first busbar and the second busbar are within an electrically insulating busbar housing, the first busbar within a first channel in the busbar housing and the second busbar within a second channel in the busbar housing.

9. The heater of claim 5, wherein the first busbar connector and the second busbar connector extend parallel to each other along an arc-shape conforming to and disposed between the arc-shaped inner end and arc-shaped outer end of the body of the heater.

10. The heater of claim 9, wherein the first heater electrode is located closer to the arc-shaped inner end than the arc-shaped outer end, and the second heater electrode is located closer to the arc-shaped outer end than the arc-shaped inner end; or wherein the first heater electrode is located closer to the arc-shaped outer end than the arc-shaped inner end, and the second heater electrode is located closer to the arc-shaped inner end than the arc-shaped outer end.

11. The heater of claim 1, wherein the body of the heater is configured for placement adjacent a bottom surface of a susceptor assembly.

12. The heater of claim 11, wherein the body of the heater is configured for connection to a support post of the susceptor assembly.

13. The heater of claim 12, wherein the body of the heater is configured to move together with the susceptor assembly.

14. The heater of claim 12, wherein the body of the heater is configured to move independent of the susceptor assembly.

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